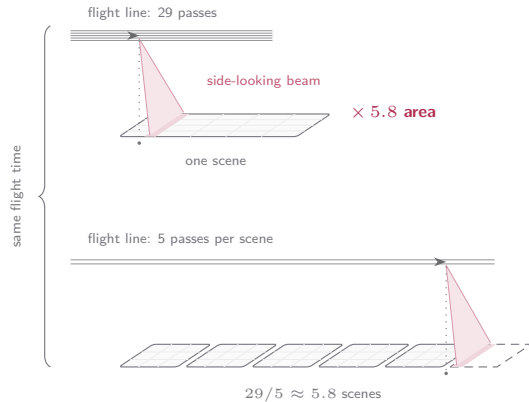
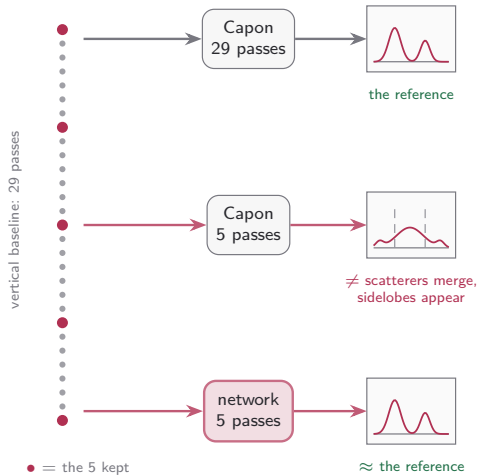


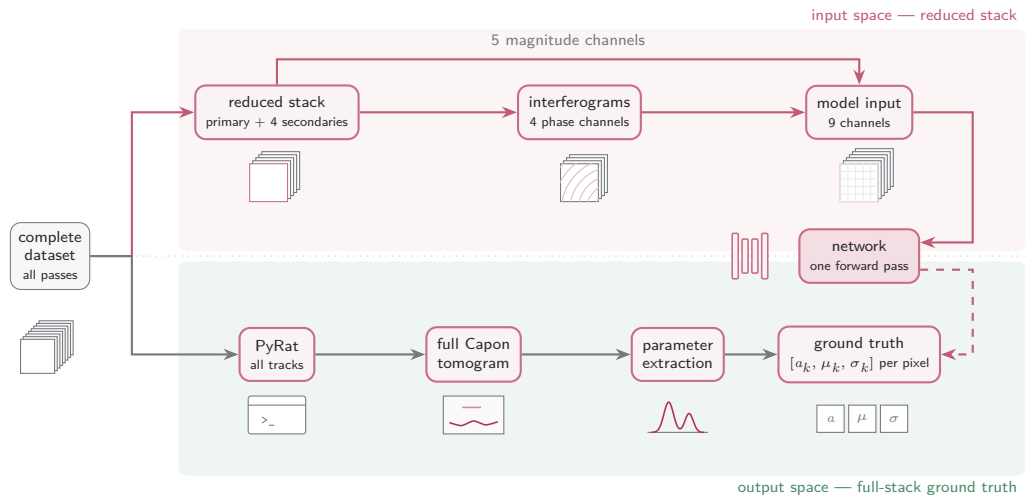
Motivation: more area from the same flight time



My contributions

- **Multi-layered output version:** the same head scaled past two scatterers per pixel ($K > 2$).
- **Multi-GPU parameter fitting on a JAX framework:** the ground-truth Gaussian fits, batched over the full scene.
- **Improvements in the model architecture:** set-prediction head, dual trunks, the backbone zoo.
- **Improvements in training stability:** matching, the leaky clamp, imbalance levers, sanity gates.
- **Improvements in pre-training:** the per-channel normalization ladder, robust statistics for the heavy-tailed channels.
- **Multiple experiments:** benchmark board, patch sweep, context ladder, input ablations, pair losses.
- **A framework to fully track training development:** launch queue, GPU scheduling, live consoles, progress and alerts.
- **A full toolset for diagnosing and visualizing results:** leaderboards, run health, cube slices, per-pixel microscope.

Methodology



A network learns to reconstruct, from the **reduced** stack alone, the per-pixel profile parameters that the full-stack tomogram and classical fit produce.

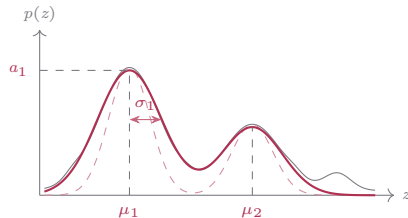
Ground truth: fitted Gaussian parameters

Supervised target: the per-pixel array $[a_k, \mu_k, \sigma_k]_{k=1}^K$ ($K=2$ today, $K=5$ in development), fit to the per-pixel max-normalised Capon spectrum $\tilde{p} = p/p_{\max}$ on the elevation grid ξ_n :

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^N (\hat{p}(\xi_n) - \tilde{p}(\xi_n))^2, \quad \hat{p}(\xi) = \sum_k a_k e^{-\frac{(\xi - \mu_k)^2}{2\sigma_k^2}}$$

One shared loss; each parameter group takes its own projected Adam step:

$$\begin{aligned} a_k &\leftarrow [a_k - \eta \text{Adam}(\partial_{a_k} \mathcal{L})]_+ \\ \mu_k &\leftarrow \text{clip}_{[\mu^-, \mu^+]} (\mu_k - \eta \text{Adam}(\partial_{\mu_k} \mathcal{L})) \\ \sigma_k &\leftarrow \text{clip}_{[\sigma^-, \sigma^+]} (\sigma_k - \eta \text{Adam}(\partial_{\sigma_k} \mathcal{L})) \end{aligned}$$



Same pixel: Capon spectrum, **peak initialisation (dashed)**, **converged fit**; the fitted $[a_k, \mu_k, \sigma_k]$ are the label.

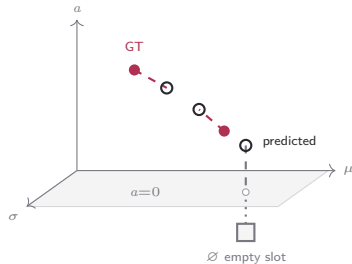
- Fitted in normalised space; the winning amplitudes are denormalised back, $a_k \rightarrow p_{\max} a_k$ (μ_k, σ_k already live on the physical axis).
- All pixels batched in one optimisation — not a per-pixel loop (next slide); most pixels activate one or two Gaussians (Act III).

Main results

- **Hungarian matching:** assigning predicted scatterers to ground-truth scatterers (next frame).
- **Set-prediction head:** gated slots that decide existence and parameters per scatterer.
- **Receptive field:** how much spatial context the network actually uses.
- **Benchmarking:** 7 backbones $\times$ 5 objectives, five seeds each: the board.
- **Dual backbone:** two trunks, one head: existence and parameters each get their own encoder.
- **Pair loss function:** param-L1 paired with a curve term.

Hungarian matching

- Each Gaussian is a **point in (a, μ, σ) space**: prediction and GT are two point sets; the matcher pairs them by distance and scores the **set**, not the slot layout.



Filled = active GT, open = predicted, $\emptyset$ = empty GT slot (below $a=0$).
Its match (grey) is zero-weighted, so $\hat{\theta}_3$ is free.

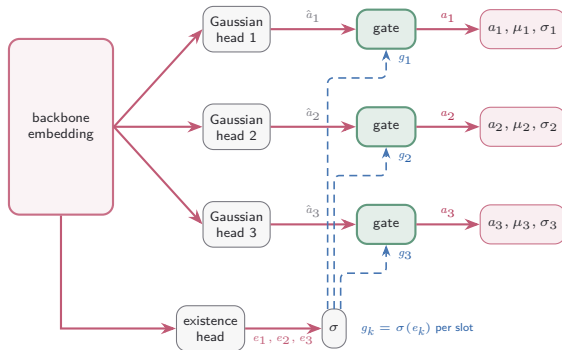
$$\mathcal{L} = \min_{\pi \in S_K} \sum_{k=1}^K w_k \left\| \hat{\theta}_{\pi(k)} - \theta_k \right\|^2 \quad \theta_k = (a_k, \mu_k, \sigma_k)$$

$w_k = 1[a_k > \tau]$ — GT *active mask*: 1 if present, 0 if empty.

	GT θ_j			
	θ_1	θ_2	$\emptyset$	
$\hat{\theta}_1$	0.9	0.2	0.0	$\xrightarrow[\text{min cost}]{\pi^*}$
$\hat{\theta}_2$	0.4	0.8	0.0	
$\hat{\theta}_3$	0.6	0.5	0.0	
cost $C_{ij} = w_j \ \hat{\theta}_i - \theta_j\ ^2$				$\sum C_{\pi^*} = 0.6$

Column $\emptyset$ is a placeholder GT ($w_j=0$) — all-zero, so the leftover prediction matches it for free: $\hat{\theta}_1 \rightarrow \theta_2$, $\hat{\theta}_2 \rightarrow \theta_1$, $\hat{\theta}_3 \rightarrow \emptyset$.

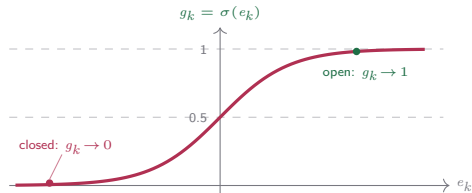
Inside the set-prediction head — gate, blend, match



each gate: $a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$
 μ_k, σ_k pass through $\cdot a_{\text{off}}$ learned per slot

The gate is a sigmoid on the existence logit e_k :

- Open slots ($g_k \rightarrow 1$): $a_k \approx \hat{a}_k$ — the raw regression survives.
- Closed slot ($g_k \rightarrow 0$): $a_k \rightarrow a_{\text{off},k} \approx 0$ whatever $\hat{a}_k$ is, collapsing onto the empty-set target at near-zero loss.
- e_k is internal — the head still emits $3K$ channels (a_k, μ_k, σ_k); a_{off} is a learned per-slot scalar (init 0).



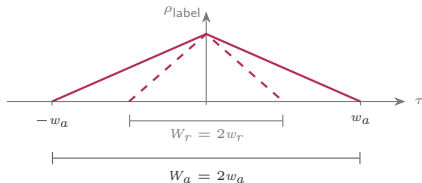
Reading the reach — the network estimates statistics

two labels at offset Δ split into shared and private looks:

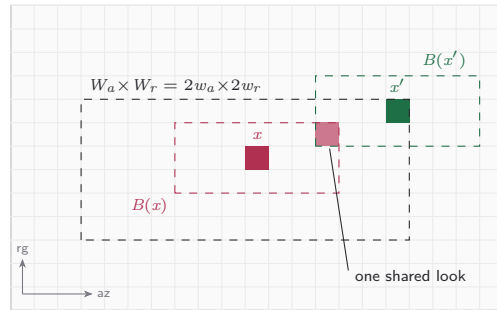
$$w_a w_r \hat{\mathbf{R}}(x) = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W} \setminus \mathcal{W}'} y_p y_p^H}_{\text{private to } x}$$

$$w_a w_r \hat{\mathbf{R}}(x') = \underbrace{\sum_{\mathcal{W} \cap \mathcal{W}'} y_p y_p^H}_{\text{shared } \mathbf{S}(\Delta)} + \underbrace{\sum_{\mathcal{W}' \setminus \mathcal{W}} y_p y_p^H}_{\text{private to } x'}$$

the shared fraction at every offset τ is the correlation law:



the label correlates over $\pm w_a$ in azimuth and $\pm w_r$ in range
— the centred patch spanning both triangles is $W_a \times W_r$.



x' sits at the far corner of the patch: $B(x')$ keeps a single shared look with $B(x)$ — one step further and the windows are disjoint ($|\Delta_a| = w_a$ or $|\Delta_r| = w_r$).

The board at five seeds — the L1 family owns the top nine

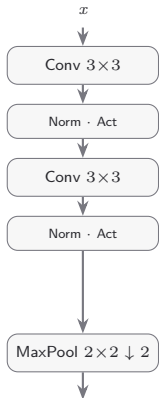
#	model	loss	curve R^2 ↑	cosine ↑	SSIM elev ↑	SSIM range ↑	SSIM azim ↑	peak err ↓	recall ↑	2-sc recall ↑	score
1	unet-skip	L1-curve	0.756 ±.012	0.924 ±.007	0.9676 ±.0004	0.9500 ±.0016	0.9611 ±.0008	1.29 ±.05	0.886 ±.006	0.728 ±.013	0.983
2	unet-skip	Charbonnier	0.751 ±.012	0.928 ±.004	0.9675 ±.0007	0.9499 ±.0024	0.9608 ±.0012	1.25 ±.03	0.886 ±.004	0.727 ±.008	0.977
3	unet-skip	param-L1	0.741 ±.016	0.931 ±.001	0.9669 ±.0004	0.9513 ±.0006	0.9608 ±.0005	1.38 ±.03	0.893 ±.004	0.745 ±.008	0.963
4	resunet	param-L1	0.734 ±.013	0.927 ±.007	0.9664 ±.0003	0.9502 ±.0006	0.9601 ±.0004	1.32 ±.07	0.894 ±.003	0.746 ±.006	0.946
5	deeplab	param-L1	0.731 ±.006	0.923 ±.004	0.9663 ±.0003	0.9495 ±.0004	0.9597 ±.0003	1.35 ±.05	0.888 ±.002	0.736 ±.005	0.937
6	deeplab	L1-curve	0.731 ±.012	0.921 ±.013	0.9667 ±.0005	0.9491 ±.0016	0.9599 ±.0008	1.33 ±.07	0.873 ±.022	0.721 ±.007	0.935
7	resunet	L1-curve	0.737 ±.019	0.916 ±.007	0.9668 ±.0004	0.9488 ±.0007	0.9601 ±.0006	1.34 ±.06	0.881 ±.005	0.715 ±.007	0.930
8	resunet	Charbonnier	0.731 ±.016	0.921 ±.014	0.9668 ±.0006	0.9492 ±.0015	0.9601 ±.0006	1.31 ±.08	0.881 ±.006	0.716 ±.008	0.929
9	deeplab	Charbonnier	0.735 ±.008	0.913 ±.015	0.9668 ±.0005	0.9492 ±.0020	0.9602 ±.0008	1.62 ±.45	0.868 ±.037	0.722 ±.013	0.912
10	nafnet	Charbonnier	0.702 ±.009	0.906 ±.007	0.9645 ±.0004	0.9444 ±.0010	0.9569 ±.0004	1.48 ±.05	0.875 ±.008	0.711 ±.009	0.829
11	segformer	Charbonnier	0.688 ±.022	0.910 ±.016	0.9651 ±.0008	0.9447 ±.0028	0.9572 ±.0013	1.51 ±.20	0.866 ±.019	0.704 ±.019	0.829
12	nafnet	L1-curve	0.698 ±.007	0.908 ±.010	0.9646 ±.0005	0.9444 ±.0013	0.9568 ±.0007	1.43 ±.08	0.879 ±.005	0.714 ±.008	0.828
⋮											
16	deeplab	MSE-curve	0.732 ±.010	0.872 ±.019	0.9615 ±.0005	0.9368 ±.0011	0.9527 ±.0004	1.62 ±.08	0.869 ±.004	0.710 ±.003	0.785
20	unet	L1-curve	0.660 ±.015	0.897 ±.026	0.9649 ±.0009	0.9441 ±.0043	0.9563 ±.0018	1.51 ±.28	0.870 ±.029	0.711 ±.020	0.760
21	unet-skip	param-MSE	0.647 ±.047	0.879 ±.017	0.9626 ±.0010	0.9398 ±.0032	0.9536 ±.0016	1.59 ±.13	0.881 ±.022	0.739 ±.014	0.692

runs: model · set-pred · Hungarian · $K=2$ · hv flips · presence A · loss · patch 64×32 , five seeds; held-out band. Cells: mean ± seed std; **bold** = best mean in column (rows shown), marked only where best and worst differ by more than 5%. score = equal-weight mean of the five metric groups (curve error, variance, SSIM, cosine, peak), min-max scaled over the 35 arms; # = rank; ranks 13–15, 17–19, 22–35 on the next frame only. cosine = per-pixel profile cosine mean; peak err in elevation units; SSIM per axis vs the GT cube (denorm); recall = matched, 2-sc recall on GT $k=2$ pixels. **green** = the winner's score; **red** = the param-MSE reference cell, 21st of 35.

One encoder block: UNet vs UNet-skip vs ResUNet

UNet

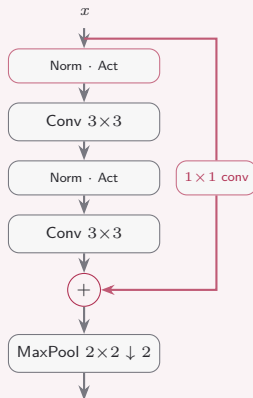
board, L1-curve: 0.760



plain double conv, no shortcut;
downsampling by max-pool

UNet-skip

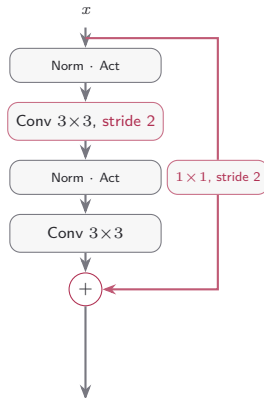
board, L1-curve: 0.983



pre-activation + **residual shortcut**;
max-pool kept

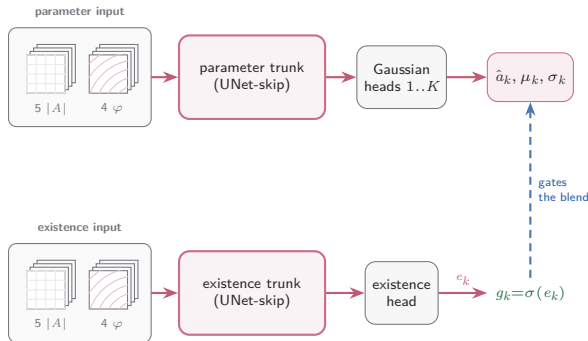
ResUNet

board, L1-curve: 0.930



downsampling inside the block: **stride-2**
conv and shortcut, no pooling (stage 1: stride 1)

The dual model — two trunks, one head



$$\text{each gate: } a_k = g_k \hat{a}_k + (1 - g_k) a_{\text{off},k}$$

the set-prediction head of Act III, unchanged — only the source of each embedding splits

- **Split the encoder, keep the head:** two independent UNet-skip trunks — the Gaussian heads read one embedding, the existence head the other. Gate blend and Hungarian matching stay as they are.
- **Each trunk picks its own stack:** the channel groups $|A|$ (5 amplitudes) and φ (4 interferograms) are selected per trunk — which modality reaches which head becomes an architectural choice.
- **Why route:** the input ablation split the labour — phase fires the slots, the full stack calibrates the parameters. The dual model can hand each head the input its job needs.

Pairing the objective — param-L1 plus a curve term

$$\mathcal{L}(w) = (\mathcal{L}_{\text{param-L1}} + w \mathcal{L}_{\text{curve}}) / (1 + w)$$



1 baseline + 15 pair arms per K , five seeds each, at $K=2$ and $K=5$ = 160 runs

The ratio block left the $K=5$ gate under-firing at every split — more existence capacity did not wake it. This experiment changes the objective instead: keep param-L1 and add a synthesized-curve term.

- **Two signals, one loss:** param-L1 reaches a slot only through its matched pair; the curve term re-synthesizes the profile from *all* slots and is permutation-invariant — its gradient touches every slot, matched or not.
- **Magnitude or shape:** L1 and Charbonnier price the synthesized profile's magnitude; cosine normalizes both curves first and prices shape alone.
- **A normalized pair:** the two terms are averaged, so the validation loss changes meaning with w — arms compare on held-out metrics only, never on the loss.
- **The question:** if the conservative gate is starved of a profile-level error signal rather than capacity, a small w should move detection at $K=5$ and leave $K=2$ alone.

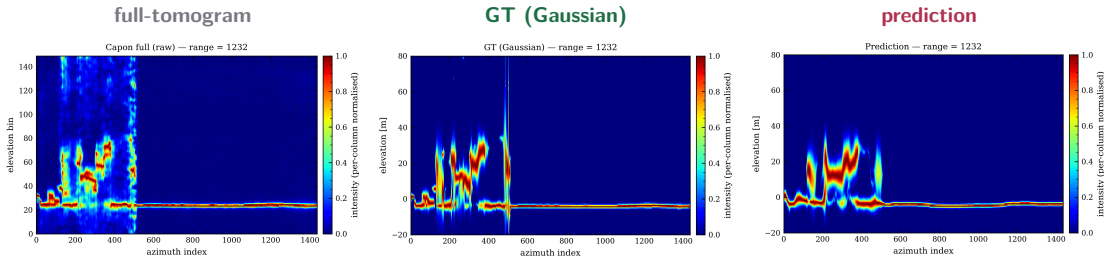
The pair winner, measured: param-L1 + 0.5 cosine

metric	value
<i>training (validation split)</i>	
best val loss ↓	0.1565 ±.0010
best epoch (val)	47
<i>reconstruction (curve)</i>	
curve RMSE ↓	0.214 ±.004
curve MAE ↓	0.0364 ±.0002
curve R^2 ↑	0.754 ±.008
per-pixel R^2 (mean) ↑	0.289 ±.011
PSNR (dB) ↑	52.76 ±.15
profile cosine ↑	0.937 ±.002
<i>structure (vs GT, denorm)</i>	
SSIM elev ↑	0.9675 ±.0002
SSIM range ↑	0.9518 ±.0002
SSIM azim ↑	0.9616 ±.0002
<i>peak location</i>	
peak err mean ↓	1.205 ±.005
peak err p95 ↓	4.70 ±0

metric	value
<i>detection (matched)</i>	
recall ↑	0.902 ±.002
$k=1$	0.999 ±.000
$k=2$	0.764 ±.005
precision ↑	0.891 ±.008
$k=1$	0.930 ±.012
$k=2$	0.827 ±.004
F1 ↑	0.897 ±.004
count exact ↑	0.905 ±.010
<i>component error (matched pairs)</i>	
μ MAE ↓	1.691 ±.036
$k=1$	0.345 ±.007
$k=2$	3.763 ±.085
σ MAE ↓	0.765 ±.005
$k=1$	0.199 ±.003
$k=2$	1.635 ±.017
a MAE ↓	0.182 ±.003
$k=1$	0.059 ±.002
$k=2$	0.372 ±.004

run: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · hv flips · presence A · full stack → both trunks · loss = param-L1 + 0.5 cosine curve; five seeded replicas, held-out band, cells mean ± seed std. SSIM per axis vs the GT cube (denorm), at 4 decimals (seed spreads ≤ 0.0004). k = true scatterer count of the pixel; μ , σ and peak error in elevation units, a in linear amplitude; recall/precision/F1 matched.

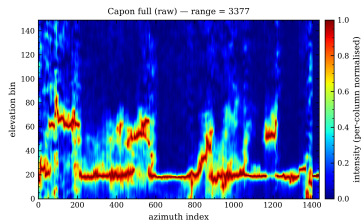
What the network reconstructs: range cut 1232



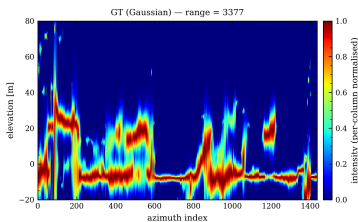
Constant-range cut $rg = 1232$ through the cube: azimuth on the horizontal axis, elevation on the vertical, every azimuth column normalised to its own peak. **Left:** the raw full-stack Capon tomogram the labels are fitted from, elevation in inversion-grid bins. **Middle:** the Gaussian mixture re-synthesised from the GT parameters, $z \in [-20, 80]$ m. **Right:** the same mixture re-synthesised from the pair winner's predicted parameters (param-L1 + 0.5 cosine).

What the network reconstructs: range cut 3377

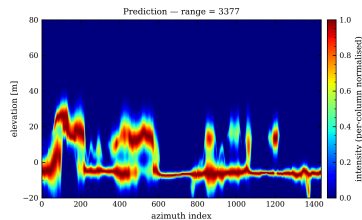
full-tomogram



GT (Gaussian)



prediction



The second cut, $rg = 3377$, same three panels: raw full-stack tomogram in inversion-grid bins, GT re-synthesis and prediction re-synthesis in $z \in [-20, 80]$ m, per-column peak normalisation.

Next steps

- **Physical loss terms:** the five ranked terms and the first single-term $K=2$ sweep results on the next six frames; the Capon-cycle arm is still queued.
- **γ -net adaptation:** adapting the γ -net architecture to this pipeline.
- **JEPA pretraining:** predicting in representation space, the closing frame.

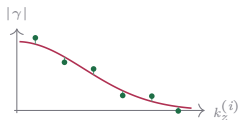
Data-consistency terms (the ranked top two)

Coherence re-synthesis

coherence domain

$$\gamma_P(k_z^{(i)}) = \frac{\sum_n P(\xi_n) e^{jk_z^{(i)} \xi_n}}{\sum_n P(\xi_n)}$$

$$\ell_{\text{coh}} = \left\langle \frac{1}{N_s} \sum_i |\gamma_P(k_z^{(i)}) - \gamma_T(k_z^{(i)})|^2 \right\rangle$$



γ_P from the predicted mixture pulled onto the target coherences,
baseline by baseline.

Cloude 2006 (PCT); same Fourier kernel as gamma-Net
(Qian et al.).

Covariance matching

covariance domain

$$\mathbf{R}[P] = \mathbf{A} \text{diag}(P) \mathbf{A}^H \Delta \xi$$

$$\ell_{\text{cov}} = \left\langle \frac{\|\mathbf{R}[P] - \mathbf{R}[T]\|_F^2}{\|\mathbf{R}[T]\|_F^2} \right\rangle$$



Synthesised vs target covariance — structure and relative scale,
entry by entry.

COMET (Ottersten, Stoica, Roy 1998); TomoSAR: Cros &
Ferro-Famil 2025; CS fidelity: Berenger et al. 2023.

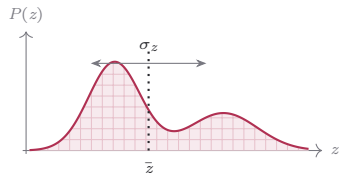
Descriptor terms: moments and total power

Moments

mass · phase-centre $\bar{z}$ · spread σ_z

$$m = \sum_n P(\xi_n) \Delta\xi$$

$$\bar{z} = \frac{\sum_n \xi_n P(\xi_n)}{\sum_n P(\xi_n)}, \quad \sigma_z^2 = \frac{\sum_n (\xi_n - \bar{z})^2 P(\xi_n)}{\sum_n P(\xi_n)}$$



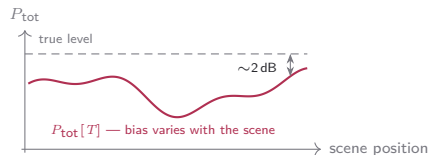
Centroid and spread are bias-robust *ratios* (Cros & Ferro-Famil 2025).

Total power

the one absolute-power quantity

$$P_{\text{tot}}[\cdot] = \sum_n (\cdot)(\xi_n) \Delta\xi$$

$$\ell_{\text{pow}} = \left\langle (P_{\text{tot}}[P] - P_{\text{tot}}[T])^2 \right\rangle$$

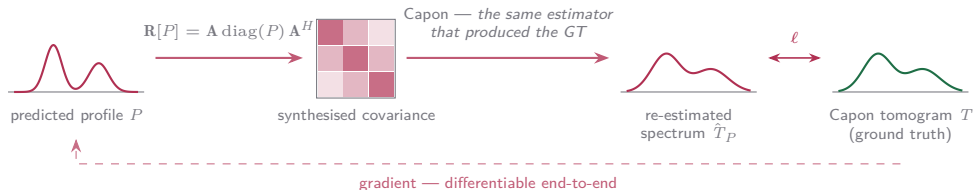


The target $P_{\text{tot}}[T]$ is not the true power: its bias moves with scene content (Lopez-Dekker & Mallorqui 2010) — the loss would imprint that bias.

The estimator-cycle term: Capon cycle

Push the prediction through the *same estimator* that produced the ground truth:

$$\hat{T}_P(\xi_n) = (\mathbf{a}^H(\xi_n)(\mathbf{R}[P] + \epsilon\bar{\sigma}\mathbf{I})^{-1}\mathbf{a}(\xi_n))^{-1}$$



- Absorbs the Capon bias and the PSF **by construction**.
- Costs an $N_s \times N_s$ solve per pixel — the most expensive of the five.

Cycle-consistency lineage: Yaman 2020, Lu 2021, Vella & Mota 2020, Angermann 2023.

The physics sweep at $K=2$ — every term rides low, the scale terms collapse high

objective	reconstruction		detection (matched)						
	curve $R^2 \uparrow$	cos med $\uparrow$	prec. $\uparrow$	recall $\uparrow$	F1 $\uparrow$	exact $\uparrow$	under $\downarrow$	over $\downarrow$	peak p95 $\downarrow$
param-L1 + cos	0.747 $\pm$.006	0.965 $\pm$.004	0.895 $\pm$.008	0.901 $\pm$.001	0.898 $\pm$.004	0.908 $\pm$.007	0.041 $\pm$.003	0.051 $\pm$.009	4.7 $\pm$.0
+ coherence re-synthesis									
$w=0.01$	0.752 $\pm$.004	0.966 $\pm$.002	0.895 $\pm$.006	0.902 $\pm$.001	0.898 $\pm$.003	0.910 $\pm$.002	0.040 $\pm$.004	0.050 $\pm$.005	4.7 $\pm$.0
$w=0.05$	0.752 $\pm$.006	0.966 $\pm$.001	0.893 $\pm$.005	0.902 $\pm$.001	0.898 $\pm$.002	0.911 $\pm$.003	0.039 $\pm$.003	0.051 $\pm$.004	4.7 $\pm$.0
$w=0.25$	0.755 $\pm$.007	0.966 $\pm$.002	0.892 $\pm$.009	0.903 $\pm$.002	0.897 $\pm$.004	0.909 $\pm$.005	0.038 $\pm$.005	0.053 $\pm$.010	4.7 $\pm$.0
$w=0.5$	0.748 $\pm$.009	0.965 $\pm$.003	0.886 $\pm$.013	0.903 $\pm$.001	0.895 $\pm$.007	0.911 $\pm$.010	0.032 $\pm$.006	0.057 $\pm$.014	4.7 $\pm$.0
$w=1$	0.751 $\pm$.008	0.966 $\pm$.002	0.870 $\pm$.045	0.903 $\pm$.001	0.886 $\pm$.024	0.885 $\pm$.063	0.032 $\pm$.005	0.083 $\pm$.068	4.7 $\pm$.0
+ covariance matching									
$w=0.01$	0.752 $\pm$.001	0.965 $\pm$.002	0.896 $\pm$.005	0.901 $\pm$.001	0.898 $\pm$.003	0.906 $\pm$.004	0.043 $\pm$.001	0.051 $\pm$.005	4.7 $\pm$.0
$w=0.05$	0.748 $\pm$.001	0.966 $\pm$.002	0.899 $\pm$.005	0.902 $\pm$.001	0.900 $\pm$.002	0.909 $\pm$.004	0.043 $\pm$.002	0.048 $\pm$.005	4.7 $\pm$.0
$w=0.25$	0.738 $\pm$.004	0.966 $\pm$.002	0.896 $\pm$.006	0.899 $\pm$.002	0.898 $\pm$.003	0.902 $\pm$.008	0.045 $\pm$.002	0.053 $\pm$.008	4.7 $\pm$.0
$w=0.5$	0.727 $\pm$.011	0.965 $\pm$.002	0.900 $\pm$.001	0.898 $\pm$.002	0.899 $\pm$.001	0.907 $\pm$.003	0.046 $\pm$.003	0.047 $\pm$.001	4.7 $\pm$.0
$w=1$	0.723 $\pm$.006	0.965 $\pm$.001	0.899 $\pm$.005	0.899 $\pm$.002	0.899 $\pm$.002	0.903 $\pm$.003	0.047 $\pm$.004	0.050 $\pm$.006	4.7 $\pm$.0
+ moments									
$w=0.01$	0.753 $\pm$.003	0.966 $\pm$.002	0.895 $\pm$.006	0.902 $\pm$.001	0.898 $\pm$.003	0.910 $\pm$.004	0.040 $\pm$.005	0.050 $\pm$.006	4.7 $\pm$.0
$w=0.05$	0.755 $\pm$.002	0.966 $\pm$.002	0.896 $\pm$.005	0.902 $\pm$.001	0.899 $\pm$.002	0.910 $\pm$.004	0.041 $\pm$.004	0.049 $\pm$.005	4.7 $\pm$.0
$w=0.25$	0.751 $\pm$.002	0.965 $\pm$.002	0.897 $\pm$.004	0.901 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.003	0.044 $\pm$.000	0.049 $\pm$.004	4.7 $\pm$.0
$w=0.5$	0.750 $\pm$.004	0.966 $\pm$.002	0.898 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.001	0.909 $\pm$.001	0.044 $\pm$.002	0.047 $\pm$.003	4.7 $\pm$.0
$w=1$	0.746 $\pm$.005	0.966 $\pm$.001	0.897 $\pm$.003	0.901 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.003	0.043 $\pm$.001	0.049 $\pm$.003	4.7 $\pm$.0
+ total power									
$w=0.01$	0.752 $\pm$.003	0.965 $\pm$.002	0.896 $\pm$.005	0.902 $\pm$.001	0.899 $\pm$.002	0.910 $\pm$.004	0.041 $\pm$.004	0.049 $\pm$.005	4.7 $\pm$.0
$w=0.05$	0.751 $\pm$.005	0.966 $\pm$.001	0.897 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.002	0.908 $\pm$.002	0.044 $\pm$.001	0.048 $\pm$.002	4.7 $\pm$.0
$w=0.25$	0.747 $\pm$.003	0.965 $\pm$.001	0.898 $\pm$.003	0.900 $\pm$.001	0.899 $\pm$.001	0.908 $\pm$.004	0.044 $\pm$.003	0.049 $\pm$.004	4.7 $\pm$.0
$w=0.5$	0.737 $\pm$.015	0.965 $\pm$.002	0.900 $\pm$.005	0.899 $\pm$.002	0.899 $\pm$.003	0.905 $\pm$.006	0.048 $\pm$.004	0.047 $\pm$.006	4.7 $\pm$.0
$w=1$	0.724 $\pm$.020	0.963 $\pm$.005	0.902 $\pm$.003	0.898 $\pm$.005	0.900 $\pm$.001	0.907 $\pm$.003	0.049 $\pm$.007	0.044 $\pm$.004	4.8 $\pm$.3

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Row = loss (param-L1 + 0.5 cos + w phys) / (1.5 + w), one physics term per arm; param-L1 + cos = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px. Val losses do not compare across arms (different composites); curve MAE / PSNR / SSIM stay flat outside the collapse arms (next frames); peak p95 is pinned at 4.7. **bold** = best in column, marked only where best and worst differ by more than 5%. **Green**: coherence $w=0.25$ takes the best curve R^2 ; at $w=1$ its gate trades misses for false alarms (under 0.041 $\rightarrow$ 0.032, over 0.051 $\rightarrow$ 0.083; seed lottery: precision $\pm$.045, exact $\pm$.063). **Red** = absolute-scale collapse past $w=0.25$ (covariance, total power). cos med: median per-pixel cosine to GT: peak p95: 95th pct of the per-pixel peak error.

The $K=2$ physics by scatterer count — the scatterer set is untouched

metric	physics term at its weight					
	none	coh. $w=0.25$	coh. $w=1$	mom. $w=0.05$	cov. $w=0.05$	pow. $w=0.05$
precision $\uparrow$	0.895 $\pm$.008	0.892 $\pm$.009	0.870 $\pm$.045	0.896 $\pm$.005	0.899 $\pm$.005	0.897 $\pm$.003
$k=1$	0.935 $\pm$.010	0.932 $\pm$.011	0.902 $\pm$.067	0.937 $\pm$.006	0.938 $\pm$.006	0.938 $\pm$.003
$k=2$	0.828 $\pm$.006	0.826 $\pm$.006	0.818 $\pm$.009	0.828 $\pm$.006	0.834 $\pm$.003	0.831 $\pm$.004
recall $\uparrow$	0.901 $\pm$.001	0.903 $\pm$.002	0.903 $\pm$.001	0.902 $\pm$.001	0.902 $\pm$.001	0.900 $\pm$.001
$k=1$	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000	0.999 $\pm$.000
$k=2$	0.763 $\pm$.002	0.766 $\pm$.004	0.768 $\pm$.001	0.764 $\pm$.003	0.763 $\pm$.003	0.761 $\pm$.003
<i>count acc pred k</i>						
pred $k=1 \uparrow$	0.944 $\pm$.004	0.948 $\pm$.006	0.954 $\pm$.002	0.944 $\pm$.005	0.943 $\pm$.002	0.941 $\pm$.001
pred $k=2 \uparrow$	0.814 $\pm$.024	0.808 $\pm$.023	0.755 $\pm$.122	0.817 $\pm$.015	0.819 $\pm$.015	0.819 $\pm$.007
μ MAE $\downarrow$	1.71 $\pm$.07	1.72 $\pm$.06	1.81 $\pm$.12	1.69 $\pm$.04	1.65 $\pm$.02	1.67 $\pm$.03
$k=1$	0.36 $\pm$.01	0.35 $\pm$.00	0.35 $\pm$.00	0.35 $\pm$.01	0.35 $\pm$.01	0.35 $\pm$.00
$k=2$	3.78 $\pm$.14	3.82 $\pm$.13	4.02 $\pm$.26	3.75 $\pm$.09	3.66 $\pm$.05	3.72 $\pm$.08
σ MAE $\downarrow$	0.774 $\pm$.011	0.778 $\pm$.011	0.809 $\pm$.020	0.771 $\pm$.018	0.754 $\pm$.007	0.761 $\pm$.005
$k=1$	0.193 $\pm$.002	0.192 $\pm$.003	0.194 $\pm$.006	0.195 $\pm$.003	0.194 $\pm$.002	0.194 $\pm$.003
$k=2$	1.668 $\pm$.023	1.675 $\pm$.016	1.737 $\pm$.035	1.657 $\pm$.043	1.620 $\pm$.017	1.639 $\pm$.015
a MAE $\downarrow$	0.182 $\pm$.004	0.182 $\pm$.005	0.185 $\pm$.006	0.182 $\pm$.003	0.187 $\pm$.005	0.184 $\pm$.004
$k=1$	0.058 $\pm$.003	0.058 $\pm$.004	0.061 $\pm$.008	0.058 $\pm$.003	0.061 $\pm$.004	0.059 $\pm$.004
$k=2$	0.373 $\pm$.006	0.373 $\pm$.005	0.373 $\pm$.004	0.372 $\pm$.002	0.381 $\pm$.007	0.377 $\pm$.004

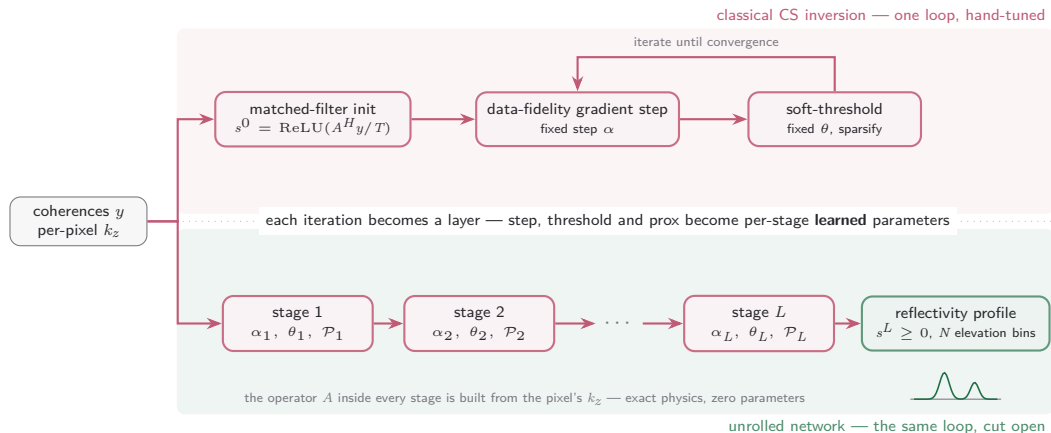
runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = physics term at its best-scoring weight, plus the $w=1$ coherence arm; none = the no-physics reference. All rows matched (Hungarian pairing); cells mean $\pm$ seed std over five seeded replicas; held-out test; k = true scatterer count of the pixel; component errors over matched pairs only. **bold** = best in row, marked only where best and worst differ by more than 5% — almost nothing clears it: detection, count calibration and component errors all sit inside combined seed noise, so the physics terms shape the profile, not the scatterer set. The one exception is full-weight coherence: its wider gate pays μ/σ (μ 1.71 $\rightarrow$ 1.81, σ 0.774 $\rightarrow$ 0.809) and destabilises the claimed pairs (pred $k=2$ acc $\pm$.122). Precision rewards under-prediction — rank on recall. count acc | pred k : share of pixels claiming exactly k scatterers whose true count is k . μ , σ in elevation units; a in linear amplitude.

The $K=2$ physics per-pixel tail — descriptor and covariance terms cap the blow-ups

metric	physics term at its weight					
	none	coh. $w=0.25$	coh. $w=1$	mom. $w=0.05$	cov. $w=0.05$	pow. $w=0.05$
<i>curve level</i>						
curve R^2 $\uparrow$	0.747 $\pm$.006	0.755 $\pm$.007	0.751 $\pm$.008	0.755 $\pm$.002	0.748 $\pm$.001	0.751 $\pm$.005
curve MAE $\downarrow$	0.0370 $\pm$.0005	0.0366 $\pm$.0003	0.0371 $\pm$.0007	0.0365 $\pm$.0002	0.0365 $\pm$.0002	0.0366 $\pm$.0001
RMSE $\downarrow$	0.217 $\pm$.002	0.213 $\pm$.003	0.215 $\pm$.003	0.214 $\pm$.001	0.217 $\pm$.000	0.215 $\pm$.002
PSNR [dB] $\uparrow$	52.63 $\pm$.10	52.78 $\pm$.13	52.70 $\pm$.14	52.76 $\pm$.04	52.65 $\pm$.02	52.70 $\pm$.08
<i>SSIM vs GT</i>						
elev $\uparrow$	0.9671 $\pm$.0004	0.9674 $\pm$.0002	0.9670 $\pm$.0006	0.9674 $\pm$.0002	0.9675 $\pm$.0002	0.9674 $\pm$.0001
range $\uparrow$	0.9510 $\pm$.0008	0.9516 $\pm$.0004	0.9499 $\pm$.0027	0.9519 $\pm$.0004	0.9521 $\pm$.0004	0.9519 $\pm$.0004
azim $\uparrow$	0.9610 $\pm$.0006	0.9615 $\pm$.0003	0.9606 $\pm$.0014	0.9615 $\pm$.0003	0.9616 $\pm$.0002	0.9615 $\pm$.0003
<i>per-pixel R^2 — the tail</i>						
median $\uparrow$	0.888 $\pm$.007	0.891 $\pm$.006	0.870 $\pm$.049	0.888 $\pm$.009	0.889 $\pm$.009	0.889 $\pm$.007
mean $\uparrow$	-1.17 $\pm$.79	-0.88 $\pm$.53	-3.71 $\pm$ 4.34	+0.53 $\pm$.19	+0.77 $\pm$.01	+0.73 $\pm$.05
p_5 $\uparrow$	0.275 $\pm$.065	0.284 $\pm$.021	-0.912 $\pm$ 2.630	0.355 $\pm$.008	0.394 $\pm$.003	0.377 $\pm$.012
p_1 $\uparrow$	-14.88 $\pm$ 11.38	-12.37 $\pm$ 4.01	-50.17 $\pm$ 70.98	-2.16 $\pm$ 1.08	-0.45 $\pm$.06	-0.87 $\pm$.35
min $\uparrow$	-13.2k $\pm$ 3.3k	-12.2k $\pm$ 4.8k	-22.5k $\pm$ 11.5k	-4.4k $\pm$ 1.2k	-4.2k $\pm$ 0.2k	-4.1k $\pm$ 0.1k
<i>per-pixel cosine</i>						
median $\uparrow$	0.965 $\pm$.004	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.002	0.966 $\pm$.001
mean $\uparrow$	0.934 $\pm$.003	0.936 $\pm$.001	0.936 $\pm$.001	0.935 $\pm$.001	0.934 $\pm$.001	0.934 $\pm$.001
p_1 $\uparrow$	0.588 $\pm$.011	0.599 $\pm$.009	0.601 $\pm$.007	0.584 $\pm$.006	0.574 $\pm$.007	0.579 $\pm$.008

runs: dual UNet-skip trunks · set-pred · Hungarian · $K=2$ · presence A · full norm ladder · hv flips · patch 64×32 · fixed routing (full stack $\rightarrow$ both trunks) · parity twins. Column = physics term at its best-scoring weight, plus the $w=1$ coherence arm; none = the no-physics reference. Five seeded replicas per arm, cells mean $\pm$ seed std; held-out test, 5.0 M px; pixel R^2 min and its spread in thousands (k). **bold** = best in row, marked only where best and worst differ by more than 5%. The typical pixel never moves (pixel R^2 median 0.888–0.891, cos med 0.965–0.966, SSIM in the fourth decimal on all three axes) — the story is the tail: a handful of catastrophic pixels wreck the no-physics mean (-1.17, p_1 -14.9, min -13.2k), and the descriptor and covariance terms cap the blow-ups (covariance $w=0.05$: mean +0.77, p_1 -0.45, min -4.2k) while cutting the curve-metric seed spread up to $\sim 5 \times$. Coherence is NOT tail insurance — at $w=1$ its tail is worse than the baseline (mean -3.71, min -22.5k): the Fourier-domain term shapes coherences, not pixel-domain outliers. The curve-level and SSIM blocks carry the headline claim: the gains that exist sit at the edge of combined seed noise.

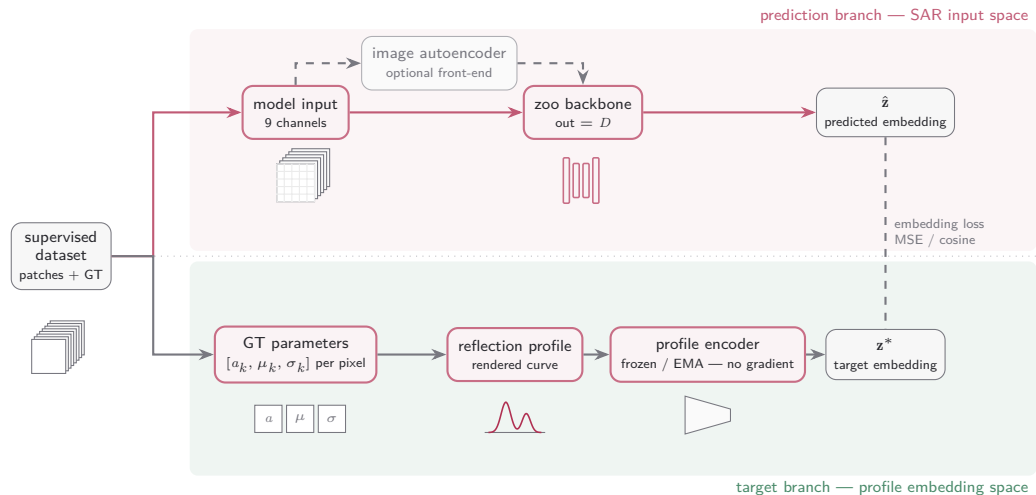
The unrolled model: the solver becomes the network



$$r^\ell = s^\ell + \alpha_\ell \operatorname{Re}[A^H(y - As^\ell)]/L_{\text{lip}}, \quad s^{\ell+1} = \operatorname{ReLU}(\mathcal{P}_\ell(r^\ell) - \theta_\ell)$$

γ -Net lineage (Qian et al., IEEE TGRS 2022); LISTA (Gregor & LeCun 2010); ISTA-Net (Zhang & Ghanem 2018). $\mathcal{P}_\ell$: per-pixel residual 1D convolution along elevation; $L_{\text{lip}} = TN \mathrm{d}z^2$ normalises the step. Default $L=8$: $\sim 1.4\text{k}$ parameters.

What JEPA does



A network learns to predict, from the reduced stack alone, the **embedding** that a separately-trained profile autoencoder assigns to the ground-truth reflection profile.